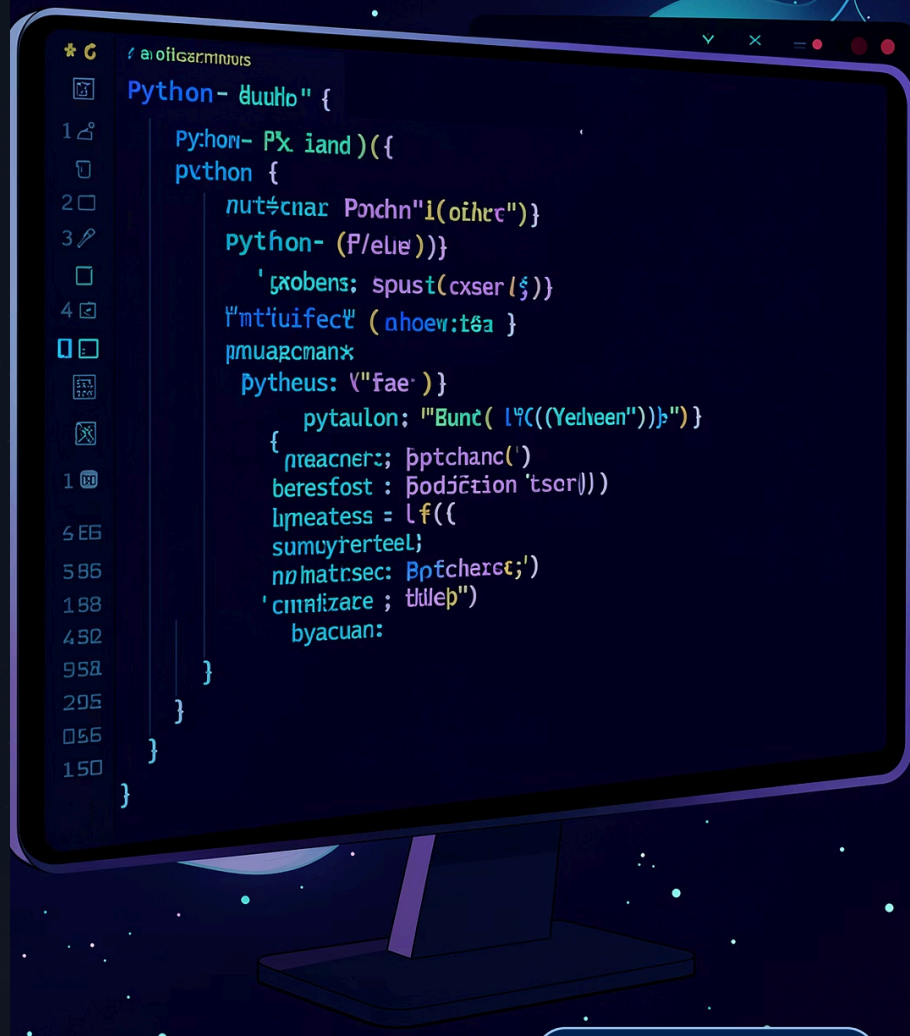
Mastering Python: Control Flow, Loops, and Functions

A visual guide to the core building blocks that make Python programs dynamic, powerful, and expressive.



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The Foundation: Making Decisions with Control Flow

By default, Python executes code line by line — but **control flow** lets your program branch, choose, and adapt based on conditions.

if Statements

Execute a block of code only when a condition evaluates to `True`.

elif & else

Chain multiple conditions and define a fallback when none are met.

match Statement

A modern, pattern-matching alternative for comparing a value against multiple cases cleanly.

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if, elif, else: The Power of Choice

```
score = 85

if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
elif score >= 70:
    grade = "C"
else:
    grade = "F"

print(f"Your grade: {grade}")
# Output: Your grade: B
```

How it works

- Python checks the `if` condition first — if `True`, that block runs.
- Each `elif` is checked in order — only the first matching block executes.
- The `else` block is a safe default — it runs when no condition above is met.

Conditions use comparison operators: `==`, `!=`, `>`, `<`, `>=`, `<=`

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